

Exercise 1. A number is divisible by 101. Prove that if you alternately add and subtract its digits in groups of two (from right to left), the result is also divisible by 101. Show that the converse is also true. For example, for 123456, you calculate $56 - 34 + 12 = 34 \neq 0$ and hence (by the converse statement), $101 \nmid 123456$.

Exercise 2. Find the remainder when 2^{2025} is divided by 11.

Exercise 3. Find the smallest positive x such that the three congruences hold: $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$ and $x \equiv 4 \pmod{9}$.

Exercise 4. Find all integers x that satisfy the congruence $5x \equiv 12 \pmod{23}$.

Exercise 5. If p is a prime show that $xy \equiv 0 \pmod{p}$ then $x \equiv 0 \pmod{p}$ OR $y \equiv 0 \pmod{p}$. Further if p is not a prime, then we can find x, y with $x \not\equiv 0 \pmod{p}$ AND $y \not\equiv 0 \pmod{p}$.

Exercise 6. Find a positive integer that leaves a remainder of 1 when divided by 2, a remainder of 2 when divided by 3, a remainder of 3 when divided by 4, ..., and a remainder of 2024 when divided by 2025,

Exercise 7. A coin collector has a large number of ancient coins, but they've never counted them all. They know that when they arrange the coins in groups of 5, there are 4 left over. When they arrange them in groups of 7, there are 6 left over. When they arrange them in groups of 11, there are 10 left over. What is the smallest possible number of coins the collector could have?

Exercise 8. Show that $(a - b)(b - c)(c - a) \equiv 0 \pmod{3}$.

Exercise 9. Find the largest positive integer that always divides the product $n(n^2 - 1)(n^2 - 4)$ for any integer $n \geq 2$.

Exercise 10. Let $a_n = 2^{2^n} + 1$. Find $\gcd(a_n, a_m)$ for $n \neq m$.